

Research Assignment 1

Home User

This computer will primarily be used for

- Word processing software.
- Home accounting software (TurboTax, Quicken).
- Children's games.
- Multimedia software for editing pictures and creating video DVDs.

What I'm looking for

Since this use case is so common, I assumed the price range would be the lowest. I searched for PCs that would accommodate a light work load and have great user-interface and ease of use. The reason for this is that most people buying computers for this use case are usually not looking for configurable or advanced devices, and are instead looking for something that is easy to set up and use.

Ideal specifications

CPU — For the CPU, I will look for a 2-core (no more than 4) since two cores is usually all it takes to run light applications for office/home work. As for threads, I ideally want a multithreaded CPU (4 threads on 2 cores) to make running multiple apps at once feel faster.

Graphics — I am not concerned with graphics very much since most integrated graphics units in CPUs nowadays can run light games. With that being said, integrated graphics is important.

Memory — Since the description didn't mention anything about heavy multitasking or budget, I feel 8GBs of DDR4 RAM is plenty for this use case as it provides enough space for the CPU to comfortably run most intended applications at a reasonable speed.

My Pick

Optiplex 3040 SFF Desktop PC, Intel Core i3-6100 3.7GHz, 8GB RAM, 256GB SSD, USB 600M WiFi, Win11 Pro

<https://www.bestbuy.com/product/dell-optiplex-3040-sff-desktop-pc-intel-core-i3-6100-3-7ghz-8gb-ram-256gb-ssd-usb-600m-wifi-win11-pro-black/JXZSQFV8Y2> (\$150)

Display: <https://www.bestbuy.com/product/msi-pro-mp251-e2-24-ips-lcd-fhd-120hz-1ms-adaptive-sync-monitor-with-hdr-readybuilt-in-speakers-displayport-hdmi--matte-black/J3P7TXSVWV> (\$85)

For a total of: **\$305**

The display that I decided to go with (\$85) I feel is a great option since its a decent size (25"), a great refresh rate for this price point (120Hz), and decent resolution (1920x1080p).

Pros

- Core i3 is perfectly fine for this use case, and can run simple/light software.
- RAM is 8GB which is what I was hoping to find.
- iGPU is fine for running simple 3d games, especially if you turn down the graphics settings or play at 720p (unlikely that you would need to unless the game is newer)
- CPU is a 2-core 4-thread which is great for this use case and exactly what I was hoping to find.
- Price is great (\$150 refurbished in excellent condition but I also found it for ~\$220 new) ~\$300 total with display and peripherals.
- 256GBs of storage which is good but not great. Would have preferred 512.
- Comes with Windows 11 Pro which is great for ease of use and set up.

Cons

- RAM is DDR3 which is slower than what I was hoping to find (DDR4)
- It's not an all-in-one which is what I was really hoping to find since it would make set-up much easier.
- It's at the limit of its use case. In that I mean if you wanted to push it past what it was meant to do, you would not find that it keeps up.

Accountant

This PC will primarily be used for:

- Office software (Microsoft office)
- Statistical software (SPSS or SAS)
- Downloading large amounts of financial data.

What I'm looking for

For this PC, I had to do some research as to what the ideal specifications would be for running SPSS and SAS. I found out that the minimum requirements were very low, but the recommended requirements jumped up in quality and speed (and price) considerably. I decided to go with something mid range but no more than \$900.

Ideal Specifications

CPU — At least an Intel core i5 or equivalent, but finding at i7 at this price point is not rare.

Graphics — I am very much not concerned with graphics for this use case since the book mentioned nothing about gaming or rendering, so I assume the iGPU will be plenty.

Memory — At least 16GBs of DDR4 (ideally DDR5) RAM since the user will almost certainly be multitasking frequently.

Storage — At least 1TB of storage since the user will be downloading large data sets for financial analysis. Ideally, I would like to find a 2TB SSD for this PC.

Display — I would prefer a wide display for this use since it will make using multiple apps easier. A wider display gives the user ample room to tile multiple windows at once. I would like a 1440p display for this user since it will make reading small numbers/text easier. 60hz is a fine refresh rate so I will look for nothing more.

My pick

Acer - Aspire All-in-One 27 FHD (AMD Ryzen 7 5825U, 16GB DDR4, 1TB PCIe SSD, AMD Radeon, Win 11 Pro)

<https://www.bestbuy.com/product/acer-aspire-all-in-one-27-fhd-amd-ryzen-7-5825u-16gb-ddr4-1tb-pcie-ssd-amd-radeon-win-11-pro-w-usb-hub-black/JJGHT4439Z>

I had to make some sacrifices for this pick to keep the price under \$1000. Namely with the display resolution, size, and storage.

For a total of: **\$931.24**

Pros

- Great CPU (Ryzen 7 5825U). Very much able to handle most software.
- 16GBs of RAM which is good for multitasking apps.
- DDR4 RAM which is very good for this use case.
- A terabyte of storage which is what I said I needed as the minimum.
- Overall good for home office use. Will handle the accountant work load just fine. My only recommendation would be to invest in a 2TB external hard drive (\$70-80) for archiving data that is no longer being used. A TB, in my opinion, will be enough for keeping currently used data on the PC for easy access.

Cons

- Smaller display than I wanted.
 - Full HD (1920x1080), I wanted at least 1440p.
 - Less than ideal storage space.
 - Not DDR5.
-

Architect

This PC will be primary used for:

- Computer Aided Drafting software like AutoCAD

What I'm looking for

I looked into what the recommended computer specs are for AutoCAD and I was surprised at how fast of a PC you need to comfortably run the software. Since AutoCAD is the most popular CAD software, I will use it as a benchmark for the PC I pick.

Ideal Specifications

CPU — Intel i7-I9 or Ryzen equivalent.

GPU — Geforce RTX 4070 with 12GBs.

Memory — At least 32 GBs of DDR4 (ideally DDR5 but current prices for DDR5 RAM are very inflated).

Storage — At least 1TB of SSD storage.

Display — 4k display, not all that concerned with refresh rate but 144 would be nice for general smoothness.

My Pick

Since this user needs a high-end workstation, I decided to configure an Origin M-Class V2 Workstation. Here are the specification I decided on (copied from the Origin PC website summery of my build):

Case : SLIGER M-CLASS

Exterior Color : Black

Processors : Intel Core Ultra 7 265K 3.9GHz (up to 5.5GHz TurboBoost)

Motherboard : MSI PRO B860-P WiFi

Memory : Crucial Pro 32GB (2x16GB) DDR5 6400MT/s

System Cooling : CORSAIR iCUE LINK TITAN 360 RX RGB AIO

System Fans : CORSAIR RX iCUE LINK RX120 Fans

Graphics Cards : NVIDIA GeForce RTX 5070

Operating System : MS Windows 11 Pro

Operating System Drive : Corsair MP600 CORE XT R2 1TB

Hard Drive : Corsair MP600 CORE XT R2 1TB

Hard Drive : 1

Power Supply : CORSAIR RM850x SERIES PLUS GOLD PSU

Display: 34 Inch Curved UltraWide™ QHD 1440P IPS HDR 10 Monitor with Dual Controller & OnScreen Control (\$299.99)

https://www.lg.com/us/monitors/lg-34wq60c-b-ultrawide-monitor?gclid=aw.ds&gad_source=1&gad_campaignid=22344465256

For a total of: **3,571**

I decided on the Intel Ultra 7 instead of the Ultra 9 because the 7 is perfectly capable for this case and the 9 would have been even more overkill.

The graphics card is a GeForce RTX 5070 which is a gaming graphics card. I purposely chose a gaming card instead of a workstation card like the Nvidia Blackwell or Pro 6000 series because while the workstation cards are made precisely for this case, they are very much overkill for AutoCAD or similar CAD software. Not to mention the \$3,000+ price difference for these cards.

Pros

- Generally very powerful, any more power would probably be for AI engineers or taxing simulations.
- Great amount of DDR5 memory (32GBs)
- Will almost definitely handle whatever alternative use case the user could want, including gaming.
- Great motherboard with a WiFi chip.
- Display is 1440p and 34" ultra-wide curved.
- Plenty of SSD space as well as slower storage for archiving.

Cons

- A little overkill.
- Very pricey but you get what you pay for.

Price Difference Analysis.

I was surprised by how cheap the first PC was. However, this is to be expected since technology is constantly being miniaturized and optimized which increases performance and decreases price. The mid-range PC is what I was expecting for the most part, but I feel I could have found a more powerful PC for that price range if I was looking for a difference use case.

The workstation for the architect I feel is very versatile in that it can function as anything from a powerful workstation to a gaming PC that can run just about any game.

#1 Super Computer

The current holder of the number one spot on the top500.org list is El Capitan, a supercomputer held in Lawrence Livermore National Laboratory in Livermore, California.

El Captain uses 4 AMD MI300A accelerator chips per node. Each of these chips has 128GBs of HBM (high-bandwidth memory), which serves as the main RAM for the chip. This means that each node has 512 GBs of memory. The super computer is made up of 11,136 total nodes, which means it has 44,544 APUs and 5.7 petabytes of HBM.

Each MI300A accelerator chip contains 24 ZEN4 CPU cores, making the total core count of the computer 1,069,056.

Each ZEN4 CPU is comparable to a Ryzen 9 7950X, only running at slower clock speeds to optimize sustained work load (1.8-3.7 GHz vs 4.5-5.7 GHz).

These computers are barely comparable to personal computers or even mainframe computers given their incomprehensible speed and power. For example, a physics simulation that can be run in 1-2 hours on the El Capitan would take a mainframe computer (IBM z16) 1-10 years to do the same.